

How to Use GPDISC — a practical guide in plain English

Version 1.1.0 - 3 September 2026 - Private and local.

GPDISC is a private consultation system that lives entirely on this computer. You type a question in your own words — exactly as you would say it to your family doctor — and it replies with a structured consultation record: what it thinks might be going on, what must not be missed, what to answer next, and what should send you back for help.

Everything stays on this computer. Nothing you type, and no answer or record GPDISC keeps, is ever sent to any external service or website. There is no account, no cloud, no transmission of any kind.

There is really only one rule for using it: **talk to it as you would talk to a doctor.** No commands, no special words, no jargon. Plain English describing how you feel is exactly what it wants.

In an emergency, call 999 (or your local emergency number) FIRST. Chest pain, stroke signs, heavy bleeding, a collapsed or blue child — telephone for help before you type anything. Software can wait; an emergency cannot. Never wait for software to tell you what you already know is urgent.

How to ask well

The difference between a vague reply and a sharp one is almost always the question. A doctor meeting you for fifteen minutes asks who, how long, how bad. Do the same in your message and GPDISC has something to work with.

Say who it is about, and their age. “My 14 year old daughter...” or “I’m 68...” changes what the sensible possibilities are. Age and pregnancy matter enormously to what a symptom can mean.

Say how long it has been going on. “Since breakfast”, “for three weeks”, “for months” — timescale sorts the worrying from the routine better than almost anything else.

Say how bad it is, in your own words. “I can still walk on it”, “I had to sit down”, “worst pain I’ve ever had” — that is exactly the language doctors use with each other.

Say what makes it better or worse. Coming on with stairs, easing when you sit forward, starting an hour after a tablet — these details often crack the case.

List the medicines, including ones you buy yourself. “I take warfarin”, “my doctor doubled my water tablet”, “regular ibuprofen from the chemist” — medicines cause and colour a remarkable share of symptoms.

Name the other conditions. Diabetes, kidney disease, pregnancy, anything long-term. The same symptom in a different body is a different problem.

Describe tests in everyday words. You do not need to understand a result to pass it on: “the ECG report says irregularly irregular, no P waves, rate 130”. Say what the paper said, plus how the person feels.

Asking about someone else is fine and common. Say who they are, their age, and what you saw: “My husband collapsed and now his speech is slurred.” For someone who cannot describe it themselves — a confused parent, a small child, someone dying — your observation is the consultation.

Asking about medicines works best when you say what the medicine is for and who takes it: “My mum takes ibuprofen for her arthritis and has kidney disease — should she still?” rather than “is ibuprofen safe?”

Asking to make sense of a test result — say the numbers and words exactly as written, who the test was on, and how that person feels. A result without the person is only half a question.

Asking for a second opinion — say what you were told, roughly by whom (“the hospital”, “my GP”), and what worries you about it. GPDISC will lay out its own reasoning to compare against what you heard.

When GPDISC asks you questions back — answer them in a follow-up message. Every reply can carry an “Ask next:” line. Those are not a quiz; they are the exact questions a doctor would ask next, and each answer you send sharpens the picture.

And when a question is asked too bare, GPDISC says so — it never guesses. These two replies are real, word for word:

Presenting complaint: Can I drink alcohol while taking metronidazole?

I don't have enough knowledge to assess this presentation - describe more or see a clinician.

Uncertainty: I don't have enough knowledge to assess this presentation - describe more or see a clinician.

Safety net: If symptoms worsen or new concerning features appear, seek medical review.

Validation: PASS - no inconsistencies, claims grounded.

Presenting complaint: I just don't feel right, I can't put my finger on it
 I don't have enough knowledge to assess this presentation - describe more or see a clinician.
 Uncertainty: I don't have enough knowledge to assess this presentation - describe more or see a clinician.
 Safety net: If symptoms worsen or new concerning features appear, seek medical review.
 Validation: PASS - no inconsistencies, claims grounded.

That is the honest answer of a system that would rather admit a gap than invent a fact. If you get one, do what it says: describe more — who, how long, how bad, what else — and ask again. The thirty examples ahead show how much a fuller question buys you.

How to read the reply

Every GPDISC reply is a consultation record with the same shape every time. Here is a real one, word for word, to a question about a cough:

Presenting complaint: I've had a cough for three weeks and I've lost a bit of weight
 Problem representation: 1 discriminating features extracted: cough
 Differential:
 - COVID-19 / respiratory viral illness with red flags (score 0.11)
 - Viral upper respiratory tract infection (score 0.10)
 - Pulmonary tuberculosis (score 0.06)
 - Bronchiolitis (score 0.05)
 - Influenza (score 0.05)
 Must-not-miss (retained):
 ! Acute asthma exacerbation
 ! Lung cancer
 ! Pulmonary tuberculosis
 ! Asthma exacerbation (child)
 ! Superior vena cava obstruction (SVC0)
 Ask next: loss of taste/smell | contact with case | self-limiting <1-2 weeks
 Investigations: oxygen saturations - severity gate
 Treatment: Symptom control + advice; breathlessness at rest or desaturation = emergency; consider antivirals in
 ↪ eligible high-risk groups.
 Referral: self-care with pharmacy support
 Safety net: Measure saturations if unwell; breathlessness at rest, blue lips or confusion = 999.
 Ruleset: United Kingdom - guidelines NICE / CKS (see guideline index); emergency number 999; notification per
 ↪ UKHSA notifiable diseases (Health Protection Regulations 2010).
 Uncertainty: Competing hypotheses remain close - treat the differential as genuinely open and use targeted
 ↪ tests to separate them. (low confidence differential - the leading match is weak; describe more or see a
 ↪ clinician)
 Validation: PASS - no inconsistencies, claims grounded.

Part by part:

- **Presenting complaint** — the question as GPDISC understood it. Check it read you right.
- **Problem representation** — the features it extracted from your words. When this line says “EMERGENCY pattern matched”, the rest of the reply is about speed, not diagnosis.
- **Differential** — the conditions it is actually considering, most likely first, each with a confidence score. A small score means a weak match, not a small problem.
- **Must-not-miss** — the dangerous conditions it refuses to forget even though they are unlikely. This list is deliberately cautious: a safety net, not a verdict that you have them.
- **Ask next** — the questions a doctor would ask you next. Answer them in a follow-up message.
- **Investigations** — the tests that would settle it, and what each is for.
- **Treatment** — what could be done now, and on whose advice.
- **Referral** — who should see you, and how soon. With the safety net, this is where the level of concern lives: “self-care with pharmacy support” is routine; “same-day urgent review” means today; “EMERGENCY ... call 999” means now.
- **Safety net** — what should change your mind and send you back for help, and how fast. Read this line even if you read nothing else.
- **Ruleset** — which country’s rules the answer ran under: guidelines, emergency number, reporting duties. Say where you are for advice that fits.
- **Uncertainty** — how sure it is, in plain words. “Competing hypotheses remain close” is an honest invitation to describe more.
- **Validation** — its own final self-check before showing you the answer. If a line here says an escalation was raised, believe the more urgent version.

One habit ties it together: read the **safety net** and **referral** lines first, then work upwards.

Thirty examples

Every answer below is a **real, unedited GPDISC reply** to the question shown, exactly as it came back from the live system. Longer replies were cut where the page ran out; the line "..." marks the cut, and nothing before it was changed. Read the questions as lessons in asking well as much as answers in their own right.

Describing your own symptoms

1. I've had a cough for three weeks and I've lost a bit of weight

Presenting complaint: I've had a cough for three weeks and I've lost a bit of weight

Problem representation: 1 discriminating features extracted: cough

Differential:

- COVID-19 / respiratory viral illness with red flags (score 0.11)
- Viral upper respiratory tract infection (score 0.10)
- Pulmonary tuberculosis (score 0.06)
- Bronchiolitis (score 0.05)
- Influenza (score 0.05)

Must-not-miss (retained):

- ! Acute asthma exacerbation
- ! Lung cancer
- ! Pulmonary tuberculosis
- ! Asthma exacerbation (child)
- ! Superior vena cava obstruction (SVC)

Ask next: loss of taste/smell | contact with case | self-limiting <1-2 weeks

Investigations: oxygen saturations - severity gate

Treatment: Symptom control + advice; breathlessness at rest or desaturation = emergency; consider antivirals in
 ↳ eligible high-risk groups.

Referral: self-care with pharmacy support

Safety net: Measure saturations if unwell; breathlessness at rest, blue lips or confusion = 999.

Ruleset: United Kingdom - guidelines NICE / CKS (see guideline index); emergency number 999; notification per
 ↳ UKHSA notifiable diseases (Health Protection Regulations 2010).

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2. I keep getting headaches every weekend, always around one eye

Presenting complaint: I keep getting headaches every weekend, always around one eye

Problem representation: 1 discriminating features extracted: headache

Differential:

- Giant cell arteritis (score 0.10)
- Tension-type headache (score 0.10)
- Meningitis / meningococcal disease (score 0.09)
- Carbon monoxide poisoning (score 0.09)
- Migraine (score 0.08)

Must-not-miss (retained):

- ! Severe hypertension
- ! Subarachnoid haemorrhage
- ! Meningitis / meningococcal disease
- ! Encephalitis
- ! Acute angle-closure glaucoma
- ! Pre-eclampsia
- ! Malaria (falciparum) - fever after travel
- ! Enteric fever (typhoid) - fever after travel
- ! Malaria (P. vivax / P. ovale)
- ! Yellow fever (vaccine-preventable haemorrhagic fever)
- ! Leptospirosis (Weil disease)
- ! Viral haemorrhagic fever (suspect: Lassa/Ebola/Marburg)
- ! Carbon monoxide poisoning
- ! Eclampsia
- ! Heat exhaustion

Ask next: age > 50, new headache | jaw ache on chewing | bilateral pressing, no photo/phonophobia, no nausea

Investigations: ESR/CRP - supportive (can be normal); temporal artery ultrasound/biopsy - confirm (after
 ↳ steroids started)

Treatment: Suspected GCA: start prednisolone 40-60 mg (or IV methylprednisolone if visual symptoms) SAME DAY
 ↳ and refer urgently; do not delay steroids for biopsy.

Referral: emergency department / 999

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3. My right knee hurts when I go down stairs

Presenting complaint: My right knee hurts when I go down stairs

Problem representation: 1 discriminating features extracted: knee_pain

Differential:

- Knee osteoarthritis (score 0.53)

Treatment: Exercise + weight loss core; topical NSAID; paracetamol; consider physio; arthroplasty referral if
 ↳ persistent functional impact (NICE NG226 OA).

Referral: routine GP review
 Safety net: A single hot, swollen, painful knee - same-day review (septic arthritis until excluded).
 Ruleset: United Kingdom - guidelines NICE / CKS (see guideline index); emergency number 999; notification per
 ↳ UKHSA notifiable diseases (Health Protection Regulations 2010).
 Uncertainty: Leading diagnosis is favoured but premature closure is a known error - dangerous alternatives
 ↳ retained below must be actively excluded.
 Validation: PASS - no inconsistencies, claims grounded.

4. I twisted my ankle playing football and it's swollen

Presenting complaint: I twisted my ankle playing football and it's swollen
 Problem representation: 1 discriminating features extracted: limb_injury_event
 Differential:
 - Closed limb fracture (score 0.29)
 - Open (compound) fracture (score 0.29)
 Ask next: pain + swelling + bony tenderness + inability to weight-bear (Ottawa ankle/knee rules) | deformity or
 ↳ crepitus is fracture until X-rayed | any wound communicating with the fracture site - even a tiny
 ↳ puncture over a deformity
 Investigations: X-ray - Ottawa rules gate the need
 Treatment: Immobilise, elevate, analgesia, same-day fracture clinic / A&E; open wounds or neurovascular change
 ↳ are emergencies.
 Referral: same-day urgent review | escalation raised routine -> urgent: ranked leader Closed limb fracture
 ↳ (urgent)
 Safety net: Numbness, pins-and-needles, or cold pale digits beyond the injury - emergency now, not tomorrow.
 Ruleset: United Kingdom - guidelines NICE / CKS (see guideline index); emergency number 999; notification per
 ↳ UKHSA notifiable diseases (Health Protection Regulations 2010).
 Uncertainty: Competing hypotheses remain close - treat the differential as genuinely open and use targeted
 ↳ tests to separate them.
 Validation:
 [BLOCK] escalation_consistency: escalation 'routine' contradicts the ranked differential (Closed limb fracture
 ↳ (urgent))
 ...

5. I feel dizzy when I stand up ever since my doctor doubled my water tablet

Presenting complaint: I feel dizzy when I stand up ever since my doctor doubled my water tablet
 Problem representation: 2 discriminating features extracted: dizziness, dizzy_on_standing
 Differential:
 - Adrenal crisis (Addisonian) (score 0.07)
 - Haemorrhagic shock (score 0.07)
 - Carbon monoxide poisoning (score 0.06)
 - Postpartum haemorrhage (PPH) (score 0.05)
 - Adverse drug effect in polypharmacy (score 0.04)
 Must-not-miss (retained):
 ! Stroke / TIA
 ! Sepsis
 ! Carbon monoxide poisoning
 ! Postpartum haemorrhage (PPH)
 ! Heat exhaustion
 Ask next: known Addison/long-term steroids missed | hyponatraemia + hyperkalaemia + hypoglycaemia | pale, cold,
 ↳ clammy with fast weak pulse after injury or bleeding - blood pressure is normal until >30% lost (a late
 ↳ sign)
 Investigations: short Synacthen test - confirms (after hydrocortisone if crisis)
 Treatment: Suspected crisis: call 999; IM hydrocortisone 100 mg immediately + IV fluids; do not delay for
 ↳ tests.
 Referral: emergency department / 999
 Safety net: Steroid-dependent person vomiting or collapsing - emergency (needs injectable hydrocortisone).
 Ruleset: United Kingdom - guidelines NICE / CKS (see guideline index); emergency number 999; notification per
 ↳ UKHSA notifiable diseases (Health Protection Regulations 2010).
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6. I've had an itchy scaly rash in my groin for a month

Presenting complaint: I've had an itchy scaly rash in my groin for a month
 Problem representation: 2 discriminating features extracted: itchy_skin, rash_generalised
 Differential:
 - Atopic eczema (score 0.38)
 - Chickenpox (varicella) (score 0.18)
 - Zika virus infection (score 0.09)
 - African tick-bite fever (rickettsia) (score 0.06)
 - Acute schistosomiasis (Katayama fever) (score 0.05)
 Must-not-miss (retained):
 ! Suspected melanoma
 Ask next: Glass test: blanching vs non-blanching | Drowsiness and poor drinking
 Treatment: Emollients generously + topical steroid potency matched to site/age; trigger avoidance; bleach baths
 ↳ for recurrent infection; step-up plan (NICE CG57).
 Referral: self-care with pharmacy support

Safety net: Clusters of small punched-out lesions, fever, or eczema suddenly worsening and painful - same-day
 ↳ (eczema herpeticum).
 Ruleset: United Kingdom - guidelines NICE / CKS (see guideline index); emergency number 999; notification per
 ↳ UKHSA notifiable diseases (Health Protection Regulations 2010).
 Uncertainty: Competing hypotheses remain close - treat the differential as genuinely open and use targeted
 ↳ tests to separate them.
 Validation: PASS - no inconsistencies, claims grounded.

Asking about someone else

7. My husband collapsed and now his speech is slurred and his face has dropped

Presenting complaint: My husband collapsed and now his speech is slurred and his face has dropped
 Problem representation: EMERGENCY pattern matched: stroke_fast
 Differential:
 Must-not-miss (retained):
 ! stroke fast
 Treatment: Do not delay transfer for further history.
 Referral: EMERGENCY: Possible stroke - call 999 immediately (FAST). Time = brain. Call 999 now.
 Safety net: Possible stroke - call 999 immediately (FAST). Time = brain.
 Ruleset: United Kingdom - guidelines NICE / CKS (see guideline index); emergency number 999; notification per
 ↳ UKHSA notifiable diseases (Health Protection Regulations 2010).
 Uncertainty: Emergency pathway overrides diagnostic refinement.
 Validation: PASS - no inconsistencies, claims grounded.

8. My father has become confused since starting a new tablet last week

Presenting complaint: My father has become confused since starting a new tablet last week
 Problem representation: 4 discriminating features extracted: confusion, medication_change_recent,
 ↳ confusion_after_injury, new_drug_started
 Differential:
 - Adverse drug effect in polypharmacy (score 0.30)
 - Moderate/severe traumatic brain injury (score 0.29)
 - Stevens-Johnson syndrome / toxic epidermal necrolysis (score 0.26)
 - Delirium (score 0.23)
 - Encephalitis (score 0.22)
 Must-not-miss (retained):
 ! Community-acquired pneumonia
 ! Meningitis / meningococcal disease
 ! Status epilepticus
 ! Encephalitis
 ! Hyperosmolar hyperglycaemic state
 ! Hypoglycaemia
 ! Adrenal crisis (Addisonian)
 ! Hypercalcaemia (incl. malignancy-related)
 ! Sepsis
 ! Delirium
 ! Frailty decompensation
 ! Acute kidney injury (pre-renal/dehydration)
 ! Enteric fever (typhoid) - fever after travel
 ! Viral haemorrhagic fever (suspect: Lassa/Ebola/Marburg)
 ! Haemorrhagic shock
 ! Crush injury / crush syndrome
 ! Salicylate (aspirin) overdose
 ! Carbon monoxide poisoning
 ! Alcohol withdrawal delirium (delirium tremens)
 ! Serotonin syndrome
 ! Methanol / ethylene glycol poisoning
 ! Puerperal sepsis (post-birth infection)
 ! Neutropenic sepsis
 ! Stevens-Johnson syndrome / toxic epidermal necrolysis
 ! Necrotising fasciitis
 ...

9. My 14 year old daughter has stopped eating and is losing weight

Presenting complaint: My 14 year old daughter has stopped eating and is losing weight
 Problem representation: 1 discriminating features extracted: weight_loss
 Differential:
 - Anorexia nervosa (score 0.16)
 - New type 1 diabetes (score 0.14)
 - Pulmonary tuberculosis (score 0.12)
 - Hyperthyroidism / thyrotoxicosis (score 0.10)
 - Lung cancer (score 0.09)
 Must-not-miss (retained):
 ! Infective endocarditis
 ! Lung cancer

! Pulmonary tuberculosis
 ! Colorectal cancer
 ! Diabetic ketoacidosis
 ! Prostate cancer (suspected)
 ! Suspected leukaemia (urgent)
 ! Amoebic liver abscess
 ! Oral cancer (suspect)
 ! Suspected Crohn's disease
 ! HIV infection (undiagnosed / late presentation)
 ! Visceral leishmaniasis (kala-azar)
 Ask next: BMI <17.5 or rapid loss | body image distortion | young/lean patient, onset over weeks
 Investigations: BMI + ECG + U&E/FBC/osteoporosis screen - physical risk assessment
 Treatment: Assess physical risk (MEED); refer eating-disorder service - children/young people: immediate
 ↳ referral (NICE NG69); GP monitors bones, ECG, electrolytes.
 Referral: same-day urgent review
 Safety net: Fainting, heart rate under 40, or rapid weight loss - same-day medical assessment.
 ...

10. My wife is 34 weeks pregnant and hasn't felt the baby move since this morning

Presenting complaint: My wife is 34 weeks pregnant and hasn't felt the baby move since this morning
 Problem representation: 1 discriminating features extracted: pregnancy_context
 Differential:
 - Threatened miscarriage (score 0.29)
 - Eclampsia (score 0.27)
 - Incomplete / ongoing miscarriage (score 0.27)
 - Imminent birth / active labour (score 0.26)
 - Cord prolapse (score 0.18)
 Must-not-miss (retained):
 ! Imminent birth / active labour
 ! Cord prolapse
 ! Incomplete / ongoing miscarriage
 Ask next: bleeding <24 weeks with a closed cervix; many settle - but an ectopic can present identically | the
 ↳ EPU scan is what separates threatened miscarriage from missed/ectopic - clinical gestation alone cannot |
 ↳ any seizure or unresponsive collapse after 20 weeks of pregnancy, in labour, or within 6 weeks of birth -
 ↳ eclampsia until proven otherwise
 Investigations: early pregnancy unit scan + hCG trajectory - location and viability
 Treatment: Same-day EPU assessment (under 18 weeks); rest does not prevent miscarriage; anti-D if rhesus
 ↳ negative; return immediately for heavy bleeding, pain or dizziness.
 Referral: same-day urgent review | escalation raised routine -> urgent: ranked leader Threatened miscarriage
 ↳ (urgent)
 ...

Asking about medicines

11. Is it safe to take paracetamol alongside warfarin?

Presenting complaint: Is it safe to take paracetamol alongside warfarin?
 Problem representation: 1 discriminating features extracted: paracetamol_ingested
 Differential:
 - Paracetamol (acetaminophen) overdose (score 0.45)
 Investigations: paracetamol level (4h post-ingestion) + LFT/INR/creatinine - plot on the treatment nomogram;
 ↳ INR is the liver-failure tracker
 Treatment: TOXBASE/poison centre advice; N-acetyl-cysteine per nomogram (start before levels return if >8h or
 ↳ staggered); never discharge without mental-health assessment if deliberate.
 Referral: emergency department / 999 | escalation raised routine -> emergency: ranked leader Paracetamol
 ↳ (acetaminophen) overdose (emergency)
 Safety net: Emergency escalation on validation - the ranked differential demands it. Any paracetamol overdose
 ↳ - however well the person looks - needs same-day hospital assessment.
 Ruleset: United Kingdom - guidelines NICE / CKS (see guideline index); emergency number 999; notification per
 ↳ UKHSA notifiable diseases (Health Protection Regulations 2010).
 Uncertainty: Leading diagnosis is favoured but premature closure is a known error - dangerous alternatives
 ↳ retained below must be actively excluded.
 ...

12. I have type 2 diabetes and chronic kidney disease, I take metformin and my eGFR is now 28, is it still safe?

Presenting complaint: I have type 2 diabetes and chronic kidney disease, I take metformin and my eGFR is now
 ↳ 28, is it still safe?
 Problem representation: 1 discriminating features extracted: known_ckd
 Differential:
 - Advanced chronic kidney disease (stage 4-5) (score 0.46)
 Investigations: eGFR + potassium + bicarbonate + Hb; urine ACR; renal ultrasound if obstruction possible -
 ↳ the number that changes management is today's potassium

Treatment: Stage-appropriate: sick-day rules (SADMANS - suspend NSAIDs, ACE, diuretics, metformin, SGLT2i
 ↳ during any dehydrating illness), anaemia check, renal dietitian, bone mineral (calcium/phosphate/PTH),
 ↳ vaccinations, and low-clearance/dialysis planning conversations BEFORE the crash landing. Every
 ↳ prescription crosses the renal check.
 Referral: routine GP review
 Safety net: Passing much less urine, breathless lying flat, palpitations or muscle weakness - same-day (fluid
 ↳ overload and potassium). Rising itch and nausea with drowsiness - urgent review.
 Ruleset: United Kingdom - guidelines NICE / CKS (see guideline index); emergency number 999; notification per
 ↳ UKHSA notifiable diseases (Health Protection Regulations 2010).
 Uncertainty: Leading diagnosis is favoured but premature closure is a known error - dangerous alternatives
 ↳ retained below must be actively excluded.
 ...

13. Should my mum still take ibuprofen for her arthritis if she has kidney disease?

Presenting complaint: Should my mum still take ibuprofen for her arthritis if she has kidney disease?
 Problem representation: 1 discriminating features extracted: known_ckd
 Differential:
 - Advanced chronic kidney disease (stage 4-5) (score 0.46)
 Investigations: eGFR + potassium + bicarbonate + Hb; urine ACR; renal ultrasound if obstruction possible -
 ↳ the number that changes management is today's potassium
 Treatment: Stage-appropriate: sick-day rules (SADMANS - suspend NSAIDs, ACE, diuretics, metformin, SGLT2i
 ↳ during any dehydrating illness), anaemia check, renal dietitian, bone mineral (calcium/phosphate/PTH),
 ↳ vaccinations, and low-clearance/dialysis planning conversations BEFORE the crash landing. Every
 ↳ prescription crosses the renal check.
 Referral: routine GP review
 Safety net: Passing much less urine, breathless lying flat, palpitations or muscle weakness - same-day (fluid
 ↳ overload and potassium). Rising itch and nausea with drowsiness - urgent review.
 Ruleset: United Kingdom - guidelines NICE / CKS (see guideline index); emergency number 999; notification per
 ↳ UKHSA notifiable diseases (Health Protection Regulations 2010).
 Uncertainty: Leading diagnosis is favoured but premature closure is a known error - dangerous alternatives
 ↳ retained below must be actively excluded.
 ...

Making sense of tests

14. I've been thirsty and passing lots of urine for a week, now I'm breathing deeply, my blood gas shows pH 7.28, pCO2 3.4, bicarbonate 12

Presenting complaint: I've been thirsty and passing lots of urine for a week, now I'm breathing deeply, my
 ↳ blood gas shows pH 7.28, pCO2 3.4, bicarbonate 12
 Problem representation: EMERGENCY pattern matched: dka
 Differential:
 Must-not-miss (retained):
 ! dka
 Treatment: Do not delay transfer for further history.
 Referral: EMERGENCY: Possible DKA - emergency; diabetic decompensation. Call 999 now.
 Safety net: Possible DKA - emergency; diabetic decompensation.
 Ruleset: United Kingdom - guidelines NICE / CKS (see guideline index); emergency number 999; notification per
 ↳ UKHSA notifiable diseases (Health Protection Regulations 2010).
 Uncertainty: Emergency pathway overrides diagnostic refinement.
 Validation: PASS - no inconsistencies, claims grounded.

15. My urine test shows blood and protein and my ankles are swollen, but it doesn't sting when I wee

Presenting complaint: My urine test shows blood and protein and my ankles are swollen, but it doesn't sting
 ↳ when I wee
 Problem representation: 2 discriminating features extracted: ankle_swelling, joint_pain_ankles_hsp
 Differential:
 - IgA vasculitis (Henoch-Schönlein purpura) (score 0.19)
 - Acute heart failure (score 0.11)
 - Advanced chronic kidney disease (stage 4-5) (score 0.04)
 - Decompensated cirrhosis (score 0.04)
 Must-not-miss (retained):
 ! Decompensated cirrhosis
 Investigations: BP measurement + urinalysis at diagnosis and weekly x4-6, FBC (platelets normal), U&E, albumin
 ↳ - the kidneys are the organ at risk; platelet count separates from ITP
 Treatment: Most children need rest, simple analgesia and BP+urine monitoring; admit for abdominal or testicular
 ↳ pain, renal involvement or being unable to walk comfortably.
 Referral: same-day urgent review
 Safety net: Return immediately with severe tummy pain, swollen or painful testicles, headache or visual change,
 ↳ or if the rash spreads to the trunk; BP and urine checks weekly for a month whatever the child's outlook
 ↳ today.
 Ruleset: United Kingdom - guidelines NICE / CKS (see guideline index); emergency number 999; notification per
 ↳ UKHSA notifiable diseases (Health Protection Regulations 2010).

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16. I've had palpitations for two days, the ECG says irregularly irregular, no P waves, rate 130

Presenting complaint: I've had palpitations for two days, the ECG says irregularly irregular, no P waves, rate
 ↳ 130

Problem representation: 1 discriminating features extracted: palpitations

Differential:

- Atrial fibrillation (new/undetected) (score 0.25)
- Panic attacks / anxiety-related symptoms (score 0.14)
- Stimulant toxicity (cocaine / MDMA / amphetamine) (score 0.09)
- Hyperthyroidism / thyrotoxicosis (score 0.08)
- Tricyclic antidepressant overdose (score 0.06)

Must-not-miss (retained):

- ! Tricyclic antidepressant overdose
- ! Stimulant toxicity (cocaine / MDMA / amphetamine)

Investigations: 12-lead ECG - diagnostic

Treatment: Rate or rhythm control plus stroke-risk assessment (CHA2DS2-VASc / ORBIT) and bleeding risk per NICE
 ↳ NG196.

Referral: routine GP review

Safety net: Palpitations with chest pain, fainting or breathlessness need same-day assessment; otherwise ECG
 ↳ within days.

Ruleset: United Kingdom - guidelines NICE / CKS (see guideline index); emergency number 999; notification per
 ↳ UKHSA notifiable diseases (Health Protection Regulations 2010).

Uncertainty: Competing hypotheses remain close - treat the differential as genuinely open and use targeted
 ↳ tests to separate them.

Validation: PASS - no inconsistencies, claims grounded.

17. I've been feverish with drenching night sweats, my doctor heard a new heart murmur, and my blood culture has grown Staphylococcus aureus

Presenting complaint: I've been feverish with drenching night sweats, my doctor heard a new heart murmur, and
 ↳ my blood culture has grown Staphylococcus aureus

Problem representation: 3 discriminating features extracted: fever, night_sweats, new_murmur

Differential:

- Infective endocarditis (score 0.41)
- Pulmonary tuberculosis (score 0.19)
- Community-acquired pneumonia (score 0.14)
- COVID-19 / respiratory viral illness with red flags (score 0.10)
- Meningitis / meningococcal disease (score 0.09)

Must-not-miss (retained):

- ! Community-acquired pneumonia
- ! Lung cancer
- ! Acute appendicitis
- ! Acute cholecystitis
- ! Acute diverticulitis
- ! Acute viral hepatitis
- ! Meningitis / meningococcal disease
- ! Encephalitis
- ! Sepsis
- ! Acute pyelonephritis
- ! Measles
- ! HIV seroconversion illness
- ! Croup (laryngotracheobronchitis)
- ! Meningococcal disease (child)
- ! Giant cell arteritis
- ! Septic arthritis
- ! Pelvic inflammatory disease
- ! Suspected leukaemia (urgent)
- ! Malaria (falciparum) - fever after travel
- ! Yellow fever (vaccine-preventable haemorrhagic fever)
- ! Leptospirosis (Weil disease)
- ! Amoebic liver abscess
- ! Viral haemorrhagic fever (suspect: Lassa/Ebola/Marburg)
- ! Peritonsillar abscess (quinsy)
- ! Epiglottitis (adult)
- ! Epididymitis (STI-associated, <35y)
- ! Infected wound

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Travel and the wider world

18. I'm going trekking in Nepal for three weeks, what should I do to prepare?

Presenting complaint: I'm going trekking in Nepal for three weeks, what should I do to prepare?

Problem representation: 1 discriminating features extracted: prep_request

Differential:

- HIV pre-exposure prophylaxis (PrEP) candidacy (score 0.42)

Investigations: HIV Ag/Ab test, hepatitis B/C, syphilis, gonorrhoea/chlamydia NAAT baseline; renal function -

↪ PrEP needs a negative HIV test and a working kidney first; Repeat HIV test at 3 months with renal

↪ monitoring - on-PrEP safety netting

Treatment: Confirm HIV negative, screen STIs and hepatitis, check eGFR, then prescribe per national PrEP scheme

↪ - daily or event-driven. Stress adherence and 3-monthly review; condoms still prevent the rest.

Referral: routine GP review

Safety net: Fever with rash and sore throat in the weeks after a new exposure - test BEFORE starting or

↪ continuing PrEP.

Ruleset: United Kingdom - guidelines NICE / CKS (see guideline index); emergency number 999; notification per

↪ UKHSA notifiable diseases (Health Protection Regulations 2010).

Uncertainty: Leading diagnosis is favoured but premature closure is a known error - dangerous alternatives

↪ retained below must be actively excluded.

Validation: PASS - no inconsistencies, claims grounded.

19. I've come back from Kenya with a fever and a rash

Presenting complaint: I've come back from Kenya with a fever and a rash

Problem representation: 2 discriminating features extracted: fever, rash_generalised

Differential:

- Community-acquired pneumonia (score 0.14)
- Infective endocarditis (score 0.11)
- Zika virus infection (score 0.11)
- COVID-19 / respiratory viral illness with red flags (score 0.10)
- African tick-bite fever (rickettsia) (score 0.10)

Must-not-miss (retained):

- ! Pulmonary tuberculosis
- ! Acute appendicitis
- ! Acute cholecystitis
- ! Acute diverticulitis
- ! Acute viral hepatitis
- ! Meningitis / meningococcal disease
- ! Encephalitis
- ! Sepsis
- ! Acute pyelonephritis
- ! Measles
- ! HIV seroconversion illness
- ! Croup (laryngotracheobronchitis)
- ! Meningococcal disease (child)
- ! Giant cell arteritis
- ! Septic arthritis
- ! Pelvic inflammatory disease
- ! Suspected leukaemia (urgent)
- ! Malaria (falciparum) - fever after travel
- ! Yellow fever (vaccine-preventable haemorrhagic fever)
- ! Leptospirosis (Weil disease)
- ! Amoebic liver abscess
- ! Viral haemorrhagic fever (suspect: Lassa/Ebola/Marburg)
- ! Peritonitis abscess (quinsy)
- ! Epiglottitis (adult)
- ! Epididymitis (STI-associated, <35y)
- ! Infected wound
- ! Serotonin syndrome
- ! Stimulant toxicity (cocaine / MDMA / amphetamine)

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20. It's been 40 degrees on my building site and my workmate is confused and has stopped sweating

Presenting complaint: It's been 40 degrees on my building site and my workmate is confused and has stopped

↪ sweating

Problem representation: 3 discriminating features extracted: sweating, confusion, hot_dry_confused

Differential:

- Heat stroke (score 0.37)
- Delirium (score 0.23)
- Encephalitis (score 0.22)
- Hypoglycaemia (score 0.18)
- Acute coronary syndrome (ST-elevation MI) (score 0.12)

Must-not-miss (retained):

- ! Acute coronary syndrome (ST-elevation MI)
- ! Acute coronary syndrome (non-ST-elevation)
- ! Community-acquired pneumonia
- ! Meningitis / meningococcal disease
- ! Status epilepticus

! Encephalitis
! Hyperosmolar hyperglycaemic state
! Hypoglycaemia
! Adrenal crisis (Addisonian)
! Hypercalcaemia (incl. malignancy-related)
! Sepsis
! Frailty decompensation
! Acute kidney injury (pre-renal/dehydration)
! Enteric fever (typhoid) - fever after travel
! Amoebic liver abscess
! Viral haemorrhagic fever (suspect: Lassa/Ebola/Marburg)
! Salicylate (aspirin) overdose
! Carbon monoxide poisoning
! Alcohol withdrawal delirium (delirium tremens)
! Serotonin syndrome
! Methanol / ethylene glycol poisoning
! Puerperal sepsis (post-birth infection)
! Neutropenic sepsis
! Necrotising fasciitis
! Decompensated cirrhosis

Investigations: Rectal/core temp (tympanic under-reads), CK, renal function, LFTs, coagulation, glucose,
↳ electrolytes - maps the organ damage

Treatment: 999 + COOL AGGRESSIVELY NOW: shade, strip, ice-water immersion where possible (gold standard), else
↳ evaporative (mist + fan + ice packs to neck/axillae/groins). Target core <39 C within 30 minutes. Fluids
↳ carefully; avoid shivering and antipyretics (useless in heat stroke).

Referral: emergency department / 999 | escalation raised routine -> emergency: ranked leader Heat stroke
↳ (emergency)

Safety net: Emergency escalation on validation - the ranked differential demands it. Someone hot and not
↳ themselves in hot conditions - start cooling before you finish taking the history.

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21. I work on an oil rig and my colleague has had chest pain for half an hour

Presenting complaint: I work on an oil rig and my colleague has had chest pain for half an hour
Problem representation: EMERGENCY pattern matched: acs
Differential:
Must-not-miss (retained):
! acs

Treatment: Do not delay transfer for further history.
Referral: EMERGENCY: Possible acute coronary syndrome - call 999. Call 999 now.
Safety net: Possible acute coronary syndrome - call 999.
Ruleset: United Kingdom - guidelines NICE / CKS (see guideline index); emergency number 999; notification per
↳ UKHSA notifiable diseases (Health Protection Regulations 2010).
Uncertainty: Emergency pathway overrides diagnostic refinement.
Validation: PASS - no inconsistencies, claims grounded.

22. My joints ache since a tick bite in the New Forest

Presenting complaint: My joints ache since a tick bite in the New Forest
Problem representation: 1 discriminating features extracted: eschar_tick_bite
Differential:
- African tick-bite fever (rickettsia) (score 0.32)

Investigations: rickettsia_serology - Paired serology; diagnosis usually retrospective
Treatment: Doxycycline 7 days - clinical diagnosis, treat before serology confirms.
Referral: routine GP review
Safety net: Confusion, neck stiffness or non-fading rash -> emergency.
Ruleset: United Kingdom - guidelines NICE / CKS (see guideline index); emergency number 999; notification per
↳ UKHSA notifiable diseases (Health Protection Regulations 2010).
Uncertainty: Leading diagnosis is favoured but premature closure is a known error - dangerous alternatives
↳ retained below must be actively excluded.
Validation: PASS - no inconsistencies, claims grounded.

Sensitive questions

23. Can I take the contraceptive pill if I get migraines with aura?

Presenting complaint: Can I take the contraceptive pill if I get migraines with aura?
Problem representation: 2 discriminating features extracted: headache, aura
Differential:
- Migraine (score 0.26)
- Giant cell arteritis (score 0.10)
- Tension-type headache (score 0.10)
- Meningitis / meningococcal disease (score 0.09)
- Carbon monoxide poisoning (score 0.09)

Must-not-miss (retained):
! Severe hypertension
! Subarachnoid haemorrhage

! Meningitis / meningococcal disease
 ! Encephalitis
 ! Acute angle-closure glaucoma
 ! Pre-eclampsia
 ! Malaria (falciparum) - fever after travel
 ! Enteric fever (typhoid) - fever after travel
 ! Malaria (P. vivax / P. ovale)
 ! Yellow fever (vaccine-preventable haemorrhagic fever)
 ! Leptospirosis (Weil disease)
 ! Viral haemorrhagic fever (suspect: Lassa/Ebola/Marburg)
 ! Carbon monoxide poisoning
 ! Eclampsia
 ! Heat exhaustion

Treatment: Triptan + NSAID early in attack; prophylaxis if ≥ 4 attacks/month; headache diary (NICE CG150
 ↳ headache).

Referral: self-care with pharmacy support

Safety net: Sudden 'worst ever' headache, fever, rash, drowsiness or new weakness - emergency, not migraine.
 ...

24. I had unprotected sex last night, what are my options?

Presenting complaint: I had unprotected sex last night, what are my options?
 Problem representation: 1 discriminating features extracted: unprotected_sex
 Differential:
 - Primary syphilis (chancre) (score 0.21)
 - Chlamydia trachomatis infection (score 0.19)
 - Gonorrhoea (score 0.19)
 - Epididymitis (STI-associated, <35y) (score 0.13)

Must-not-miss (retained):
 ! Gonorrhoea
 ! Epididymitis (STI-associated, <35y)

Ask next: Classically PAINLESS indurated ulcer 10-90 days post-exposure - pain argues for herpes | Regional
 ↳ painless inguinal nodes; spontaneous healing does NOT mean resolution | Majority asymptomatic - screening
 ↳ history beats symptom pattern

Investigations: syphilis_serology - Treponemal EIA first-line; RPR/VDRL titre for activity

Treatment: Benzathine penicillin G single IM dose (doxycycline if penicillin-allergic). GUM same-day; full STI
 ↳ screen; partner notification.

Referral: same-day urgent review | escalation raised routine -> urgent: ranked leader Primary syphilis
 ↳ (chancre) (urgent)

Safety net: Rash, fever or warts weeks later = secondary syphilis - return; untreated syphilis causes
 ↳ cardiovascular/neurological disease.

Ruleset: United Kingdom - guidelines NICE / CKS (see guideline index); emergency number 999; notification per
 ↳ UKHSA notifiable diseases (Health Protection Regulations 2010).
 ...

25. I think I've got an STI, there's a discharge

Presenting complaint: I think I've got an STI, there's a discharge
 Problem representation: 1 discriminating features extracted: vaginal_discharge
 Differential:
 - Pelvic inflammatory disease (score 0.13)
 - Bacterial vaginosis (score 0.04)
 - Chlamydia trachomatis infection (score 0.04)
 - Gonorrhoea (score 0.04)
 - Trichomoniasis (score 0.04)

Must-not-miss (retained):
 ! Gonorrhoea

Investigations: STI screen (chlamydia/gonorrhoea) - supportive (can be negative)

Treatment: Empirical antibiotics per BASHH: ceftriaxone IM + doxycycline + metronidazole 14 days; analgesia;
 ↳ partner notification; exclude pregnancy first (BASHH/NICE).

Referral: same-day urgent review

Safety net: Pain becoming severe, fever, vomiting, or faintness - same-day review; always treat partners.

Ruleset: United Kingdom - guidelines NICE / CKS (see guideline index); emergency number 999; notification per
 ↳ UKHSA notifiable diseases (Health Protection Regulations 2010).

Uncertainty: Leading diagnosis is favoured but premature closure is a known error - dangerous alternatives
 ↳ retained below must be actively excluded.

Validation: PASS - no inconsistencies, claims grounded.

26. I'm 25 and worried about a lump in my testicle

Presenting complaint: I'm 25 and worried about a lump in my testicle
 Problem representation: 1 discriminating features extracted: anxiety
 Differential:
 - Panic attacks / anxiety-related symptoms (score 0.50)
 - Generalised anxiety disorder (score 0.24)
 - Hyperthyroidism / thyrotoxicosis (score 0.04)
 - Alcohol dependence (score 0.03)

Treatment: Breathing retraining; explanation; CBT referral if recurrent; screen depression; avoid
 ↳ benzodiazepines long-term (NICE CG113).
 Referral: routine GP review
 Safety net: Chest pain with sweating, or fainting, or lasting >15 min is not a panic attack until physical
 ↳ causes excluded.
 Ruleset: United Kingdom - guidelines NICE / CKS (see guideline index); emergency number 999; notification per
 ↳ UKHSA notifiable diseases (Health Protection Regulations 2010).
 Uncertainty: Competing hypotheses remain close - treat the differential as genuinely open and use targeted
 ↳ tests to separate them.
 Validation: PASS - no inconsistencies, claims grounded.

Mind, memory and long-term illness

27. I've felt low since my wife died and I can't sleep

Presenting complaint: I've felt low since my wife died and I can't sleep
 Problem representation: 1 discriminating features extracted: poor_sleep
 Differential:
 - Depression (moderate) (score 0.16)
 - Generalised anxiety disorder (score 0.09)
 - Alcohol dependence (score 0.05)
 Investigations: PHQ-9 - severity + monitoring
 Treatment: PHQ-9 severity; guided self-help/low-intensity CBT first; SSRI if moderate-severe or preference
 ↳ (NICE CG90); always ask about suicidal thoughts.
 Referral: routine GP review
 Safety net: Thoughts of harming yourself - seek help same day; 999 if there is a plan or intent.
 Ruleset: United Kingdom - guidelines NICE / CKS (see guideline index); emergency number 999; notification per
 ↳ UKHSA notifiable diseases (Health Protection Regulations 2010).
 Uncertainty: Competing hypotheses remain close - treat the differential as genuinely open and use targeted
 ↳ tests to separate them.
 Validation: PASS - no inconsistencies, claims grounded.

28. My son says I've been forgetting things for months

Presenting complaint: My son says I've been forgetting things for months
 Problem representation: 3 discriminating features extracted: poor_concentration, memory_problems,
 ↳ progressive_forgetfulness
 Differential:
 - Suspected dementia (incl. Alzheimer/vascular/Lewy-body) (score 0.43)
 - Depression (moderate) (score 0.19)
 - Vitamin B12 deficiency (score 0.07)
 - First-episode psychosis (score 0.02)
 Must-not-miss (retained):
 ! First-episode psychosis
 Investigations: Reversible-cause bloods (FBC, U&E, LFT, TFT, calcium, glucose, CRP, B12/folate) - excludes
 ↳ the treatable mimics before any dementia label; Structured cognitive test (MoCA / mini-Cog / 6-CIT) -
 ↳ quantifies the deficit and tracks progression; CT head (or MRI) - excludes structural cause; not to
 ↳ diagnose dementia itself
 Treatment: Take a collateral history; review the drug list (anticholinergics, opioids, benzodiazepines); assess
 ↳ home safety (cooker, driving, medication handling); refer to a memory assessment service - do not label
 ↳ dementia in one consultation.
 Referral: routine GP review
 Safety net: Sudden confusion, drowsiness, rapid worsening or not managing fluids needs same-day assessment -
 ↳ that is delirium, not dementia, until proven otherwise.
 ...

29. My terminally ill father gets agitated every evening

Presenting complaint: My terminally ill father gets agitated every evening
 Problem representation: 1 discriminating features extracted: agitation
 Differential:
 - Stimulant toxicity (cocaine / MDMA / amphetamine) (score 0.18)
 Treatment: Emergency: active cooling, benzodiazepines in generous doses, fluids; treat cocaine chest pain as
 ↳ ACS (avoid beta-blockers).
 Referral: emergency department / 999
 Safety net: Anyone hot, agitated and twitching after stimulants needs emergency care now - cooled, not calmed
 ↳ down at home.
 Ruleset: United Kingdom - guidelines NICE / CKS (see guideline index); emergency number 999; notification per
 ↳ UKHSA notifiable diseases (Health Protection Regulations 2010).
 Uncertainty: Leading diagnosis is favoured but premature closure is a known error - dangerous alternatives
 ↳ retained below must be actively excluded.
 Validation: PASS - no inconsistencies, claims grounded.

30. My dad's Parkinson's tablets were changed and now he's seeing things

Presenting complaint: My dad's Parkinson's tablets were changed and now he's seeing things
 Problem representation: 2 discriminating features extracted: hallucination, hallucinations

Differential:

- First-episode psychosis (score 0.30)
- Alcohol withdrawal delirium (delirium tremens) (score 0.22)
- Delirium (score 0.11)

Must-not-miss (retained):

! Delirium

Investigations: physical + drug screen - exclude organic causes

Treatment: Urgent referral to early intervention in psychosis team (within 2 weeks; same-day if risk); do not
→ start antipsychotics in primary care routinely (NICE CG178).

Referral: same-day urgent review | escalation raised routine → urgent: ranked leader First-episode psychosis
→ (urgent)

Safety net: Voices or beliefs that others don't share, getting worse - urgent mental health assessment; 999
→ if there is risk to life.

Ruleset: United Kingdom - guidelines NICE / CKS (see guideline index); emergency number 999; notification per
→ UKHSA notifiable diseases (Health Protection Regulations 2010).

Uncertainty: Competing hypotheses remain close - treat the differential as genuinely open and use targeted
→ tests to separate them.

Validation:

[BLOCK] escalation_consistency: escalation 'routine' contradicts the ranked differential (First-episode
→ psychosis (urgent))

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A last word

Notice how often the replies above end with a safety net — *what should bring you back, and how fast*. That is GPDISC's most important sentence, every time.

If in doubt, get a human. A symptom that frightens you; anything a reply itself calls an emergency: **call 999 in the UK, 112 or 911 elsewhere — first, and never wait for software.**

GPDISC provides second opinions, education and support for thinking. It is **not a replacement for professional medical care**, not a diagnosis to act on alone, and not a substitute for the emergency services. All medical decisions belong with qualified clinicians, taken together with you.